Evaluating Two-Stream Symbolic Music Generation: Experiment Plan Report

July 30, 2026

Companion document to `EXPERIMENTS.md` (the operational protocol).

1 Setup

Tasks. Generation of symbolic music with two paired streams:

1. melody and chord pair. TRAINING corpus: POP909 (∼900 songs) + Nottingham (∼1,000 two-track tunes) combined, which roughly doubles the melody/chord training data; EVALUATION stays POP909-only (§1, held-out ids), so metrics remain interpretable in the target domain and Nottingham contributes training signal only;
2. drum and non-drum pair (LA corpus; evaluated on held-out RWC).

Generation modes.

1. co-generation (co-gen);
2. marginal generation (marg. gen);
3. conditional generation (cond. gen).

Baselines.

ID	Construction
S0	pretrained single-stream CP transformer, trained on the LA dataset
S1	S0 finetuned on the COMBINED POP909+Nottingham melody–chord data (merged streams), i.e. the same corpus the duet systems train on
S-mel / S-chord	S0 finetuned on *one* stream of the combined melody–chord data
Y	YinYang: frozen S0 + low-rank cross-attention adapters + LoRA; conditional generation

Our models.

ID	Model
A.1	DuetAttn (`M2CIntraCrossAttn`)
A.2	DuetBlockDiffusion (`M2CDuetBlockDiffusion`)
B.1	DuetAnticipatory (`M2CDuetAnticipatory`)
C.1	`M2CDuetRehearsal`
C.2	`M2CDuetPrefix`

2 Research questions

RQ1 — Co-generation. Does explicit two-stream joint modeling beat a merged-stream model at generating both parts together?

RQ2 — Marginals. Does joint training *retain* the ability to generate each stream alone (a non-inferiority question, not a superiority one)?

RQ3 — Conditional generation. Given the full partner stream, how does the *conditioning horizon* (no future / bounded future / unbounded future) shape quality, and do fully-trained conditional duets beat parameter-efficient adapter conditioning?

RQ4 — Mechanisms. Which architectural components are causally load-bearing: the within-frame refinement mechanism, the training objective behind it, and the MoE routing?

3 Experiment blocks

Table 1 lists, per experiment, every model that must be trained (or already exists) and the dataset it is trained on. Checkpoints are reused across experiments wherever the same system appears; only the first row that introduces a checkpoint carries a training cost.

3.1 E1 — Co-generation (headline)

Both streams are generated jointly from a 4-bar prompt of both. Corpus-matched pairing: A.2-drumnondrum vs. **S0** (both LA-trained); A.2-melchord vs. **S1** (both trained on the combined POP909+Nottingham corpus) — so the measured gap is attributable to architecture alone. B.1 participates per-stream as the competing duet strategy.

Four pre-registered hypotheses map each burden the single-stream model carries to a specific observable (Table 2).

Pre-registered counter-expectations: ties on local plausibility and pitch-class histograms (the matched baseline saw the same data); possible loss on the structure axis (A.2's known repetition tendency).

3.2 E2 — Marginal generation (guard; retention, not superiority)

Claim: joint training did not cost the model its marginals. Opponent: the per-stream *specialist* — same pretrained initialization, same finetune recipe, same songs, single-modality data — the strictest fair opponent (each devotes full capacity to one marginal). Success criterion: non-inferiority within a pre-registered margin ($\delta = 0.05$ on primary JSD grammar metrics). Outside-margin results are reported as the *quantified marginal cost* of joint training; a better-than-specialist result is reported as positive cross-stream transfer.

Mechanism note (pre-registered): in marginal mode the absent stream is *clamped* to explicit silence (never sampled), and under the frozen adaptive decode the refinement loop exits every step — marginal-mode A.2 is therefore its AR pathway. Disclosed risk: an all-silent partner is out-of-distribution for the melody/chord model (chords are essentially always active in BOTH corpora of the combined mix — Nottingham tunes carry a rendered chord track throughout, and the ∼13 melody-only tunes were excluded during preprocessing precisely so the streams stayed paired — so adding Nottingham does NOT relieve this risk), while it is in-distribution for drum/non-drum, where drum-less songs occur naturally. The two-task contrast is therefore still the built-in diagnostic separating capacity interference from distribution shift.

3.3 E3 — Conditional generation (horizon dose–response)

The full ground-truth partner is given; the model generates the other stream. The systems form an ordered conditioning-horizon spectrum (Table 3).

Scoring separates two notions of quality: *fit to the given stream* (primary — what conditioning is for; computed by the metric module on the pair ⟨given ground truth, generated⟩) and agreement with the original target (secondary only — many accompaniments are valid for one partner). Hypotheses:

H-E3.1 (sanity gate, within-family). Each system must beat its own co-generation output scored against the ground-truth partner — same model, same decode, only the conditioning differs.

H-E3.2 (dose–response). Monotone improvement along the horizon spectrum; the headline sub-question is the fraction of the C.2 − A.2 gap B.1 recovers (a large fraction $\Rightarrow$ bounded, streamable conditioning suffices).

H-E3.3 (training vs. adapters). Fully-trained duets vs. adapters at fixed unbounded horizon — deliberately registered without a directional prior, as both outcomes are informative.

3.4 E4 — Ablations

E4a — mechanism factorization. A.2 differs from A.1 in a coupled objective + decode change; three contrasts separate them: A.2($K{\geq}1$) vs. A.2($K{=}0$) isolates the decode-time refinement (weights shared — free); A.2($K{=}0$) vs. A.1 is the training-objective parity check (expected parity; deficit = objective tax, surplus = beneficial auxiliary task); A.1 vs. A.2($K{=}4$) is the combined package (expected $\approx$ additive). The refinement gain is predicted to be *metric-selective* (landing on H2, the fit block) — a uniform gain would indicate generic decode compute rather than the claimed mechanism. Disclosed confound: A.1 and A.2 are separate training runs; their step counts are recorded and any imbalance stated with its bias direction.

E4b — decode schedule (melody/chord replication). $K \in \{0, 1, 4\}$ (expected: clearer monotone H2 benefit than on drum/non-drum, since both streams are always active; a flat curve would weaken the $K{=}4$ headline choice); adaptive early-exit on/off at $K{=}4$ (expected null on melody/chord — confirming the mechanism is silence-specific); optional temperature-schedule comparison.

E4c — MoE ablation. A.2 vs. **A.2-dense** (compute-matched dense FFN), identical otherwise. Predicted deficit profile is the mirror image of E4a's: landing on H3 (grammar) and secondarily H1, with H2 least affected — the coupling is carried by the attention structure both variants share. Combined with the observational expert-routing analysis, a positive result closes the specialization story from both ends; a null is reported as MoE-not-load-bearing.

3.5 E5 — Anticipation-horizon sweep (optional, gated)

Only if E3 shows B.1 recovering a substantial fraction of the unbounded-horizon gap: train B.1 at $k \in \{8, 16, 32\}$ and plot the conditional fit against k, with A.1 as the $k{=}0$ point and C.2 as the $k{=}\infty$ asymptote — one dose–response figure locating the knee of the lookahead curve.

4 Evaluation methodology

4.1 Objective metrics (implemented: `eval_metrics.py`)

All metrics are *reference-calibrated*: distributional metrics are Jensen–Shannon divergences against the ground-truth continuation's distributions; fit metrics are deltas against the same statistic computed on the reference pair; density is a ratio plus a drift slope. Scoring is confined to the continuation window (frames 64–384). Reporting follows the pre-registered priority ordering H3 > H2 > H1, with one primary endpoint per hypothesis; the remaining metrics are supporting diagnostics. The module was validated on synthetic fixtures in which each engineered failure mode (grammar chaos, stream death) trips exactly its own hypothesis block.

4.2 Listening test

Blind pairwise A/B comparisons on three axes — (a) overall musicality, (b) inter-stream coherence, (c) 24-bar structure — with ≥ 3 raters, randomized order, and samples pre-drawn at fixed seed order (no curation). Win rates per axis with a sign test. The H4 prediction is about the *profile* across axes, not the overall win rate.

4.3 Frozen decode settings

One configuration per system, frozen before scoring: A.2 at $K{=}4$, final temperature 0.9, top-p 0.95, adaptive early-exit on ($K{=}4$ exercises the full trained refinement depth; the temperature schedule is the prior listening-test selection; the adaptive guard covers the known silent-frame regression and is itself ablated in E4b). All other systems at temperature 1.0 with their script defaults. Any change to these settings invalidates and reruns the affected grid.

4.4 Statistics

Paired per-song comparisons on identical prompts; three samples averaged per song before testing; Wilcoxon signed-rank on per-song means. With 10–15 evaluation songs, only large effects reach conventional significance — objective metrics are treated as directional evidence and the listening test carries the headline perceptual claims.

5 Threats to validity (all pre-registered)

1. **Genre mixing in the melody/chord training corpus** — POP909 (pop piano) and Nottingham (folk tunes) differ in harmonic rhythm, voicing conventions and phrase structure, so the combined corpus does not represent one homogeneous "melody/chord grammar." This is accepted deliberately: it roughly doubles a corpus small enough that the first S1 finetune attempt overfit before its first validation measurement, and EVALUATION remains POP909-only so the reported numbers stay in the target domain. The cost is that H3 (stream-appropriate grammar) is tested against a model whose training grammar is a genre mixture; a system could in principle lose H3 points for importing folk conventions into pop continuations. All melody/chord systems — duet and baseline alike — share the same mixture, so the COMPARISON stays matched even though the absolute H3 values are harder to interpret.

2. **Y-mc domain position** — the melody/chord YinYang is Nottingham-trained. It is reported as an external reference, never as a matched baseline. Note this threat CHANGED with the combined corpus: Nottingham is now part of every melody/chord system's training mix, so Y-mc is no longer the only system exposed to folk data — but it is still the only one that never saw POP909, which is the evaluation domain. A POP909-matched YinYang finetune remains the identified optional upgrade.
3. **E4a training-budget imbalance** — A.1 and A.2 are separate runs; step counts are recorded and any imbalance disclosed with its bias direction.
4. **E2 architecture non-constancy** — there is no non-degenerate "A.2 trained marginally," so E2 compares the joint system's marginal against the single-stream pipeline's best specialist; the specialist also sees half the tokens per song (an asymmetry that makes an A.2 win evidence of positive transfer, not a confound).
5. **Multiple valid accompaniments** — ground-truth agreement is never a primary conditional endpoint; fit-to-given is.
6. **Small evaluation set** — 10 POP909 + 5–15 RWC songs bounds statistical power; mitigated by the paired design, three samples per cell, and the listening test as the perceptual arbiter.
7. **Marginal-mode decode path** — under the frozen adaptive setting, E2's A.2 outputs come from the AR pathway (refinement exits on the clamped-silent partner); stated so results are attributed to the right mechanism.

6 Execution roadmap

Phase 0 — prerequisites. Data preparation for the combined melody/chord corpus is complete: Nottingham was tokenized into paired cp4 streams (melody = track 0, chords = track 1; ∼13 melody-only tunes excluded from BOTH streams so the pairing gate passes), a program-tagged single-stream variant was built for the S* baselines, and each was concatenated after POP909 — POP909 FIRST and unreordered, which preserves its songs' absolute indices and therefore keeps the held-out evaluation ids held out under the index-mod-10 split.

1. A.2 melody/chord on the combined corpus — **trained**.
2. B.1 melody/chord on the combined corpus, k=16 — **trained**.
3. S1 on the combined merged/tagged corpus — **trained**. *Status note:* the first S1 attempt (20k steps, max-LR 2×10^{-5}, validation every 250 steps) overfit before its FIRST validation measurement — ∼20 steps/epoch at this batch put that measurement ∼12 epochs in, past the minimum, and validation loss rose monotonically 0.37 → 0.74 while train loss fell to 0.10. The retuned recipe (2k-step schedule, max-LR 10^{-5}, validation every 25 steps) captured the real dip on POP909-only: 0.39 → ∼0.358 over ∼1k steps, then a clean plateau. The same short recipe is used for the combined corpus and the specialists; because the combined corpus is roughly twice the size, its validation minimum is expected LATER in step count, so a still-descending curve at 2k steps calls for a fresh 4k-step run rather than a resume (the OneCycle LR has annealed to ∼0 by then).
4. S-mel / S-chord specialists on the combined single-modality streams — **to train**.
5. E4c: A.2-dense on the combined corpus — **to train**.
6. Evaluation prompt folders for the ten held-out POP909 ids — built automatically by `eval_e1.sbatch`; RWC folders already exist.
7. Locate the Y-dn / Y-mc checkpoints. (Metric and scoring modules: complete and dry-run

verified.)

Phase 1 — generation sweeps. All inference via existing batch scripts: the duet lanes, the S*/specialist lanes, the YinYang lanes, the E4a A.1 runs (existing checkpoint), and the E4b decode grid. A.1-melchord training and the drum/non-drum specialists are *gated* on the corresponding first-task results.

Phase 2 — scoring. Manifest construction; `eval_metrics.py` over all outputs; per-mode comparison tables in the pre-registered priority order.

Phase 3 — listening test on the E1/E3 shortlist; vote collation.

Phase 4 — write-up. One results table per experiment block; the E4b metric-vs-K figure; the E4a factorization chart; the E5 dose–response figure if gated in.

Approximate additional training cost beyond runs already underway: three short finetunes (S1, S-mel, S-chord $\approx$ a few GPU-hours each) plus one full A.2-dense run ($\approx$ one A.2-melchord budget). All evaluation sweeps are single-GPU inference jobs.

7 Planned deliverables

- **T1**: E1 co-generation table (systems $\times$ H1–H3 primaries + listening win rates), per task.
- **T2**: E2 retention table (A.2 marginal vs. specialist, margin verdict per stream).
- **T3**: E3 horizon table ordered by conditioning horizon, with the B.1 gap-recovery fraction highlighted.
- **F1**: E4a factorization bar chart (decode effect / objective effect / combined).
- **F2**: E4b metric-vs-K curve.
- **F3**: expert routing by stream (observational companion to E4c).
- **T4**: E4c MoE table (A.2 vs. A.2-dense across hypothesis blocks).
- Listening-test protocol sheet and anonymized vote record.

Exp.	System	Training data	Status
E1	A.2 (drum/non-drum)	LA drum + non-drum pair (cp16)	trained (98k)
E1	A.2 (melody/chord)	COMBINED melody + chord pair (POP909 + Nottingham, cp4)	trained
E1	B.1 (drum/non-drum)	LA drum + non-drum pair	trained
E1	B.1 (melody/chord)	COMBINED melody + chord pair, k=16	trained
E1	S0	LA merged single stream (pretraining)	trained
E1	S1	COMBINED merged, program-tagged (cp16); finetune of S0	trained (short recipe)
E2	A.2	— (reuses the E1 checkpoints)	—
E2	S-mel	COMBINED melody-only stream; finetune of S0, short recipe	**to train**
E2	S-chord	COMBINED chord-only stream; finetune of S0, short recipe	**to train**
E2	S-drum / S-nondrum	LA drum-only / non-drum-only; finetunes of S0	gated on melchord E2
E3	A.2, B.1	— (reuse the E1 checkpoints)	—
E3	C.1, C.2	LA drum + non-drum pair	trained
E3	Y-dn	LA drum-subset, 31k songs (adapters + LoRA on frozen S0)	existing paper ckpt
E3	Y-mc	Nottingham, 1,020 songs (adapters + LoRA on frozen S0)	existing paper ckpt
E3	(optional domain match)	a POP909-matched YinYang finetune, or a Nottingham-only evaluation slice; Nottingham is now part of the duet TRAINING mix, so Y-mc is no longer the only system seeing folk data	optional
E4a	A.1 (drum/non-drum)	LA drum + non-drum pair	trained
E4a	A.1 (melody/chord)	COMBINED melody + chord pair	gated on drum/non-drum E4a
E4b	—	decode-time only; reuses the A.2-melchord checkpoint	—
E4c	A.2-dense	COMBINED melody + chord pair (identical recipe to A.2-melchord)	**to train**
E5	B.1 at $k \in \{8, 32\}$	LA drum + non-drum pair (k=16 exists)	gated on E3

Table 1: Training requirements by experiment. Bold = new training runs currently owed; “gated” = trained only if the triggering result warrants it.

#	Judges	Prediction	Primary metric
H1	each stream alone	consistent role & texture across 24 bars (melody monophonic, chords block-voiced, stable density, no stream death); S* shows role leakage, density drift, or stream death	`survival_min`
H2	mutual fit	chords consonant with the simultaneous melody; non-drum onsets lock to drums	`chord_tone_cov_delta` / `onset_sync_delta`
H3	grammar	stream-appropriate statistics (harmonic rhythm ≈ 1 change/bar, contour smoothness); MoE experts separate by stream	`harmonic_rhythm_jsd` / `onset_grid_jsd`
H4	listeners	an *ordering* of listening margins: coherence > musicality $\geq 0 \approx$ structure — a uniform profile would indicate generic quality, not the mechanism	pairwise A/B per axis

Table 2: E1 pre-registered hypotheses.

Horizon of visible partner future	System	Streamable
0 frames (same-frame)	A.2 conditional mode	yes
$k = 16$ frames (1 bar)	B.1	yes, k-frame latency
unbounded (bidirectional)	C.1, C.2, Y	no

Table 3: E3 conditioning-horizon spectrum.